

Avoidance of On-Target Off-Tumor Activation Using a Co-stimulation-Only Chimeric Antigen Receptor

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Chimeric antigen receptors (CARs) combine T cell activation with antibody-mediated tumor antigen specificity, bypassing the need for T cell receptor (TCR) ligation. A limitation of CAR technology is on-target off-tumor toxicity caused by target antigen expression on normal cells. Using GD2 as a model cancer antigen, we hypothesized that this could be minimized by using T cells expressing V γ 9V δ 2 TCR, which recognizes transformed cells in a major histocompatibility complex (MHC)-unrestricted manner, in combination with a co-stimulatory CAR that would function independently of the TCR. An anti-GD2 CAR containing a solitary endodomain derived from the NKG2D adaptor DAP10 was expressed in V γ 9V δ 2⁺ T cells. Differential ligation of the CAR and/or TCR using antibody-coated beads showed that pro-inflammatory cytokine response depended on activation of both receptors. Moreover, in killing assays, GD2-expressing neuroblastoma cells that engaged the V γ 9V δ 2 TCR were efficiently lysed, whereas cells that expressed GD2 equivalently but did not engage the V γ 9V δ 2 TCR were untouched. Differentiation between X-on tumor and X-off tumor offers potential for safer immunotherapy and broader target selection.

INTRODUCTION

Chimeric antigen receptor (CAR)-based immunotherapeutics have been developed to harness the cytotoxic potential of peripheral blood T cells. CARs comprise ectodomains, which recognize tumor-associated antigens, fused to endodomains, which provide co-stimulatory signals. The majority of CARs currently under investigation contain the CD3 ζ endodomain, which provides T cell receptor (TCR) signaling, in addition to one (second-generation) or two (third-generation) co-stimulatory endodomains, such as CD28, 4-1BB, or OX40.¹ When expressed in α β T cells, second- and third-generation CARs bypass the requirement for major histocompatibility complex (MHC)-restricted antigen presentation, providing CD3 ζ -mediated TCR signaling and co-stimulatory signals in the presence of the target antigen. CAR⁺ T cells targeting CD19 have shown unprecedented clinical efficacy in lymphoid malignancy in adults and children.^{2,3} Progress in targeting solid tumors, such as neuroblastoma, has been more measured, though CAR⁺ T cells are attractive as they offer the potential for sustained anti-tumor immunity. CARs targeting GD2

in neuroblastoma have shown early promise,^{4,5} but the potential for sustained neurotoxicity remains a concern, especially given the severe toxicities encountered in other CAR⁺ T cell trials as a result of unexpected off-tumor antigen expression.^{6,7} As a means to overcome this limitation, we have devised a CAR design in which signals 1 and 2 of T cell activation are provided by separate receptors.

GD2 is a ganglioside abundantly expressed on the surface of neuroblastoma cells and on several other cancer types. It is a highly ranked tumor-associated antigen and an attractive target for immunotherapy.⁸ In 2010, it was shown that adjuvant immunotherapy comprising an antibody (ch14.18-Sp2/0) targeting GD2, in combination with IL-2, GM-CSF, and isotretinoin, significantly improved the event-free and overall survival of high-risk neuroblastoma patients.⁹ The opiate requirements of patients receiving GD2-targeting antibody are substantial, and a minority of patients develop long-term neurological sequelae.^{9,10} The degree of acute and chronic neurotoxicity that will emerge from ongoing trials using second- and third-generation GD2-targeting CARs is not yet known.

Comprising 1%–5% of circulating T cell numbers, γ δ T cells possess innate and adaptive immune functions. Their ability to recognize danger-associated molecular patterns to differentiate between healthy and tumor cells in an MHC-independent manner¹¹ distinguishes them from α β T cells, which rely on MHC-restricted antigen presentation to detect altered self. The most abundant γ δ T cell subset in human peripheral blood expresses the V γ 9V δ 2 TCR (hereafter referred to as V δ 2⁺ γ δ T cells), responding to increased phosphoantigen production in target cells.¹² Phosphoantigens, such as isopentenyl-5-pyrophosphate, are cholesterol biosynthesis intermediates, and their accumulation in response to metabolic dysregulation has been linked to aberrant epidermal growth factor receptor (EGFR) activity.^{13,14} Detection of phosphoantigen by the V γ 9V δ 2 TCR is indirect, and,

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unlike antigen recognition by $\alpha\beta$ T cells, it does not depend on MHC-presented epitopes. Phosphoantigen detection is thought to occur through its binding to butyrophilin-3A1, although the mechanism by which TCR engagement is effected is a matter of intense debate, and other models have been proposed.^{15–18}

TCR engagement alone is insufficient to induce activation of, and cytotoxicity by, naive T cells, and a second co-stimulatory signal is required. For $V\delta 2^+$ $\gamma\delta$ T cell cytotoxicity against neuroblastoma, co-stimulation is known to occur in part through receptors, such as NKG2D, which are ligated by danger-associated molecules, such as MIC-A and MIC-B, but can also be provided via Fc γ receptors within the context of an opsonized target cell, as we have previously shown.^{19,20} The degree of antibody-dependent cell-mediated cytotoxicity (ADCC) was proportional to the degree of antigen-antibody-CD16 interaction.^{19,20} Neuroblastoma cells have been reported to evade NKG2D-mediated immune destruction through downregulation of NKG2D ligands or by secretion of receptor-blocking soluble ligands.^{21,22} This has been demonstrated to be an important escape mechanism against NK cell-mediated killing of neuroblastoma.²³ Therefore, the NKG2D adaptor protein DAP10 was an attractive candidate for inclusion in a co-stimulatory CAR designed to target neuroblastoma. This approach is a development of previously reported strategies that divided the CD3 ζ and co-stimulatory signals between two CARs targeting different tumor-associated antigens.^{24,25} By using the pre-existing danger-sensing specificity of the $\gamma\delta$ TCR, it is possible to escape the requirement for two distinct tumor-associated antigens.

In this report, we demonstrate that $V\delta 2^+$ $\gamma\delta$ T cells are amenable to manipulation with a CAR comprising a conventional ectodomain to target GD2 linked to an endodomain that provides DAP10 co-stimulation but no TCR signal. This design supplements but does not replace the endogenous $\gamma\delta$ TCR signal, capitalizing on the innate ability of $\gamma\delta$ T cells to differentiate between healthy and transformed self in an MHC-independent manner. Thus, co-stimulation-only CARs provide cytotoxic activation upon encountering a cancer cell, but not upon encountering non-transformed cells expressing the same target antigen.

RESULTS

Second-Generation CARs Override the Requirement of $V\delta 2^+$ $\gamma\delta$ T Cells for a Second Signal to Induce Cytotoxicity against Neuroblastoma

To mimic current immunotherapeutic strategy, we expressed a second-generation anti-GD2 CAR containing CD3 ζ and CD28 endodomains (GD2-28- ζ ; Figure 1A) in $V\delta 2^+$ cells, and we compared the antibody-dependent and -independent killing of GD2 $^+$ and GD2 $^-$ neuroblastoma cell lines. Expression of GD2-28- ζ in the $V\delta 2^+$ $\gamma\delta$ T cells was associated with increased antibody-independent cytotoxicity (Figure 1B) against GD2 $^+$ cell lines (Figures 1B and 1C). The benefit of GD2-28- ζ CAR expression on $\gamma\delta$ T cell cytotoxicity against the GD2 $^+$ cell line Lan1 was highly significant (cytotoxicity at 10:1 effector: target ratio $24.8\% \pm 4.6\%$ versus $5.4\% \pm 1.9\%$; Figure 1C), indicating that CAR expression in zoledronate-expanded $V\delta 2^+$ is

capable of transducing cytotoxic signals in these cells. It should be noted that the transduction efficiency and duration of stimulation differ between zoledronic acid-induced CAR $^+$ $\gamma\delta$ T cells and the CAR $^+$ $\alpha\beta$ T cells stimulated with anti-CD3/28 beads, though cytotoxicity assays were always performed at day 6 after transduction. Therefore, comparison of relative cytotoxic potential of these populations cannot be inferred from this experiment.

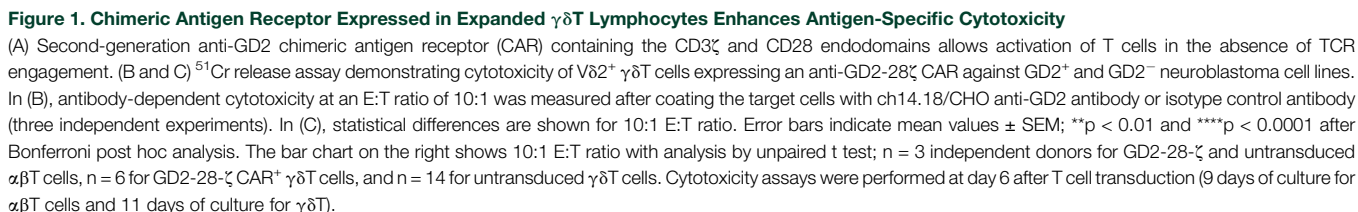
Generation and Detection of CAR-Expressing $V\delta 2^+$ $\gamma\delta$ T Cells

We speculated that it might be possible to retain full signal 2-dependent killing of neuroblastoma by $V\delta 2^+$ cells while avoiding killing of non-transformed GD2-expressing cells by exploiting the ability of the V $\gamma 9$ V $\delta 2$ TCR to provide signal 1 following engagement by stressed, but not by non-stressed, cells. To this end, we generated a CAR comprising a GD2-binding ectodomain and an endodomain containing the DAP10 endodomain, hereafter referred to as GD2-DAP10 (Figure 2A). We compared it in subsequent experiments to the previously described GD2-28- ζ CAR. Both the GD2-28- ζ and GD2-DAP10 CARs contain a linker derived from the CH2CH3 portion of human Fc, and they are co-expressed with a marker containing a minimal CD34 epitope, which stains brightly. Either can be used to determine transduction efficiency (Figure 2B). Using zoledronate-expanded $V\delta 2^+$ cells, transduction efficiencies of 22.4%–87.2% (mean $58.6\% \pm 3.7\%$) were achieved for GD2-DAP10 and 5.0%–77.3% (mean $41.5\% \pm 5.8\%$) for GD2-28- ζ (Figure 2C). Overall proportions of $V\delta 2^+$ cells following transduction varied widely between donors, but the choice of CAR construct used had no significant effect. After 10 days of expansion (day 5 after transduction), it was possible to obtain $1.75 \times 10^6 \pm 2.4 \times 10^5$ GD2-DAP10 CAR $^+$ $V\delta 2^+$ $\gamma\delta$ T cells and $1.29 \times 10^6 \pm 2.2 \times 10^5$ GD2-28- ζ CAR $^+$ $V\delta 2^+$ $\gamma\delta$ T cells per milliliter of blood drawn (Figure 2D).

Specific and Selective Activation of GD2-DAP10 CAR-Expressing $\gamma\delta$ T Cells

To investigate the specific activation of GD2-DAP10 CAR $^+$ $\gamma\delta$ T cells in comparison to GD2-28- ζ CAR $^+$ $\gamma\delta$ T cells without the influence of additional confounding factors, we established an in vitro assay using antibody-coated anti-biotin beads (Figures S1 and S2). These beads were labeled with biotinylated anti-CD3 antibody to engage the V $\gamma 9$ V $\delta 2$ TCR and/or anti-CAR antibody to engage the CH2CH3 linker in the respective CARs (Figure 3A). By coating the beads with only one of the two antibodies, $\gamma\delta$ TCR and CAR intracellular signals could be isolated from each other. Due to the changing presence or absence of either signal 1 or signal 2 in this system, differences in the cytokine production profile were expected between $\gamma\delta$ T cells expressing either the GD2-DAP10 or GD2-28- ζ CAR. Intracellular cytokine production was assessed by flow cytometry (Figure 3B), which demonstrated that, while stimulation of the CAR alone was sufficient to stimulate interferon (IFN)- γ production in GD2-28- ζ CAR $^+$ $V\delta 2^+$ $\gamma\delta$ T cells, CD3 and CAR stimulation was required to stimulate IFN- γ production in GD2-DAP10 CAR $^+$ $V\delta 2^+$ $\gamma\delta$ T cells.

Incubation with unlabelled or anti-CD3 antibody-coated beads did not induce the production of IFN- γ in $\gamma\delta$ T cells transduced with



of the CAR alone, but not by engagement of the TCR, indicating that the DAP10-derived signal 2 alone was sufficient to generate a TNF- α response (Figure 3D). Interestingly, this TNF- α was not detectable by ELISA, suggesting that CAR co-stimulatory signal might have led to an accumulation of intracellular non-secreted cytokine (Figure 4).

To further characterize $\gamma\delta T$ cell function following ligation of either the GD2-DAP10 CAR or the GD2-28- ζ CAR, we measured the concentration of cytokines in the supernatant following stimulation with beads as described above. Release of IFN- γ , TNF- α , interleukin-2 (IL-2), IL-4, and Granzyme B by GD2-DAP10 V $\delta 2^+$ cells was only seen when both CD3 and the CAR were stimulated. If CD3 or the GD2-DAP10 CAR was stimulated in isolation, cytokine

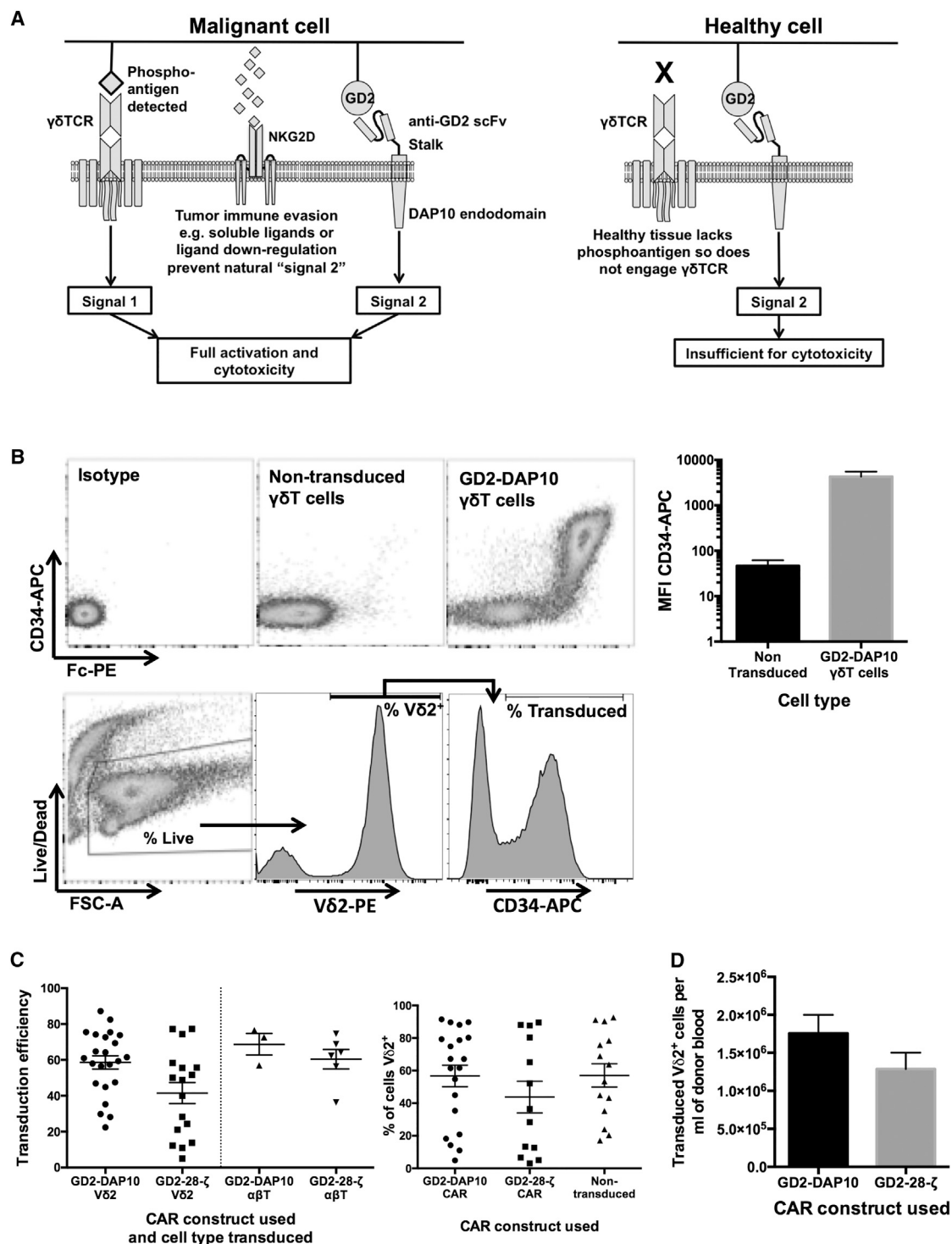


Figure 2. GD2-DAP10 CARs Expressed in Vδ2⁺ γδT Cells

(A) Vδ2⁺ γδT cells are activated in an MHC-independent manner by indirectly sensing increased phosphoantigen levels on tumors through engagement of the TCR with cognate ligand. Tumors evade γδT cell cytotoxicity by suppressing the co-stimulatory signals required. CARs containing co-stimulatory domains can bypass this evasion while retaining tumor specificity. Healthy tissues do not produce increased phosphoantigen levels so that the TCR is not engaged and the CAR alone is insufficient to provide full activation. (B) The GD2-DAP10 and GD2-28-ζ CARs have an Fc-derived linker and are co-expressed with a marker gene containing a minimal CD34 epitope. Transduced (legend continued on next page)

release was minimal or absent. This was not the case for GD2-28- ζ V δ 2⁺ cells, which, as expected, produced substantial amounts of these cytokines following CAR ligation alone. Interestingly GD2-28- ζ V δ 2⁺ cells also produced IL-10 following CAR stimulation, which was not seen in the GD2-DAP10 V δ 2⁺ cells, even when CD3 and the CAR were stimulated (Figure 4).

$\gamma\delta$ T Cells Expressing GD2-DAP10 CARs Show Cytotoxicity against GD2⁺ Neuroblastoma and Ewing Sarcoma In Vitro

To evaluate possible therapeutic efficacy of the GD2-DAP10 CAR in V δ 2⁺ $\gamma\delta$ T cells, we evaluated specific cytotoxicity against representative cell lines derived from the childhood cancers neuroblastoma and Ewing sarcoma, which we and others have previously demonstrated to express GD2 uniformly.^{19,29} Expression of the GD2-DAP10 CAR in V δ 2⁺ $\gamma\delta$ T cells yielded significantly enhanced cytotoxicity against the GD2⁺ neuroblastoma cell line LAN-1, which was equivalent to the cytotoxicity imparted by the GD2-28- ζ CAR expressed in V δ 2⁺ $\gamma\delta$ T cells (Figure 5A). This effect was also seen against GD2⁺ Ewing sarcoma cell lines, such as TC-71 (Figure 5B), but it was not seen against GD2-non-expressing neuroblastoma cell line SK-N-SH (Figure 5C). To demonstrate that the GD2-DAP10 CAR was not providing sufficient signal to elicit killing independently of the $\gamma\delta$ TCR, we expressed the same construct in $\alpha\beta$ T cells. Expression of the GD2-DAP10 CAR had no effect on $\alpha\beta$ T cell cytotoxicity against GD2⁺ neuroblastoma cell lines unlike the conventional GD2-28- ζ CAR, which demonstrated equivalent antigen-specific cytotoxicity against LAN-1 GD2⁺ neuroblastoma cells when expressed in $\alpha\beta$ T cells and V δ 2⁺ $\gamma\delta$ T cells (Figures 5A and 5D).

$\gamma\delta$ T Cells Expressing GD2-DAP10 CARs Do Not Show Toxicity against GD2⁺ Cells that Do Not Engage the $\gamma\delta$ TCR

Phosphoantigens, such as isopentenyl-5-pyrophosphate (IPP), are produced in increased quantities by cancer cells, and, in human cells, they are detected by the V γ 9V δ 2 $\gamma\delta$ TCR via mechanisms involving intermediate molecules, such as butyrophilin.¹⁵ This process has been linked to aberrant EGFR activity, which is seen in many tumor types.^{13,14} Healthy cells do not produce increased levels of phosphoantigen and, therefore, do not engage the $\gamma\delta$ TCR. When using GD2 as a model antigen, culture of healthy human GD2⁺ cells is not technically or ethically feasible, given that GD2 expression is restricted to peripheral neurons. MHC-independent phosphoantigen recognition by the V γ 9V δ 2 $\gamma\delta$ TCR is species specific;^{30,31} therefore, we developed a murine model of a GD2⁺ cell line that would not engage the human V γ 9V δ 2 $\gamma\delta$ TCR by transducing the murine colon carcinoma cell line CT26 to express GD2, as murine cells do not engage the human V γ 9V δ 2⁺ TCR (Figure 6A).

First, to demonstrate the suitability of the antigen model, we determined that the expression of GD2 alone was sufficient to increase the susceptibility of CT26 to killing by $\alpha\beta$ T cells expressing the GD2-28- ζ CAR (Figure 6B). V δ 2⁺ $\gamma\delta$ T cells expressing the second-generation GD2-28- ζ CAR retained the ability to kill CT26-GD2 cells; in contrast, GD2-DAP10 CAR⁺ V δ 2⁺ $\gamma\delta$ T cells did not significantly kill CT26-GD2 cells, consistent with failure of $\gamma\delta$ TCR engagement and lack of the necessary signal 1 (Figure 6C). The killing by GD2-DAP10 CAR⁺ V δ 2⁺ $\gamma\delta$ T cells was dependent on the presence of the target antigen, because wild-type CT26 elicited no killing response from these effectors (Figure 6C, right panel). Hence, just as a lack of engagement of the TCR in GD2-DAP10 CAR⁺ V δ 2 cells leads to the failure of IFN- γ production in the bead-based system (Figures 3C and 4), failure to engage the TCR by a cellular target results in a failure to elicit cytotoxicity.

$\gamma\delta$ T Cells Expressing GD2-DAP10 CARs Have a Favorable Phenotype for Adoptive Transfer

The ability to exert specific cytotoxicity against tumor cells must be accompanied by the ability to mount an antigen-specific secondary expansion of CAR⁺ T cell populations. Failure to do this has been linked to T cell exhaustion and obstructs the efficacy of CAR T cells used in clinical trials.³² Unfortunately, testing in vivo persistence in animal models is complicated by the potential for a xenoreponse to drive T cell expansion, which would confound results. Therefore an in vitro model of rested CAR⁺ cells was used. V δ 2⁺ cells expressing the GD2-DAP10 CAR, which had been left for 3 weeks without antigenic challenge following initial zoledronic acid stimulation, were re-challenged with an irradiated GD2⁺ cell line (LAN-1) or a GD2⁻ control (SK-N-SH) or simply maintained in a target-free state. Co-culture with the GD2⁺ neuroblastoma cell line LAN-1 led to a sustained secondary expansion over a 3-week period, whereas incubation with the GD2⁻ neuroblastoma cell line SK-N-SH produced no significant difference from cells incubated with no targets (Figure 7A).

T cell exhaustion is associated with phenotypic and functional changes. Exhausted T cells are less able to mount a secondary expansion when challenged, and they will express cell surface markers, such as TIM-3 and PD-1. Exhaustion attenuates the response of adoptively transferred CAR T cells and reduces their clinical efficacy, an effect that can be reduced through choice of CAR endodomain.³² To determine the susceptibility of $\gamma\delta$ T cells expressing either GD2-28- ζ CARs or GD2-DAP10 CARs to exhaustion, we determined their expression of TIM-3 and PD-1 over the course of a prolonged culture (Figure 7B). For the purposes of statistical analysis, exhaustion was defined as the expression of at least one of the markers TIM-3 and PD-1. Expression

cells are therefore Fc γ /CD34⁺. The gating strategy for determining transduction efficiency is shown. Transduced cells have much greater median fluorescence intensity (MFI) for CD34-APC than non-transduced cells (4,242 $\pm$ 1,215 versus 46.6 $\pm$ 15.4; n = 6). (C) Transduction efficiency of V δ 2⁺ cells and $\alpha\beta$ T cells was consistent, though there is a large degree of donor-donor variation in V δ 2⁺ cell percentages after transduction. Transduction with either CAR had no significant effect on V δ 2⁺ percentage. (D) After 10 days of expansion, we determined the total number of CAR-expressing V δ 2⁺ $\gamma\delta$ T cells and related this to the volume of blood drawn to generate this many cells (n = 8 independent donors). Error bars denote SEM.

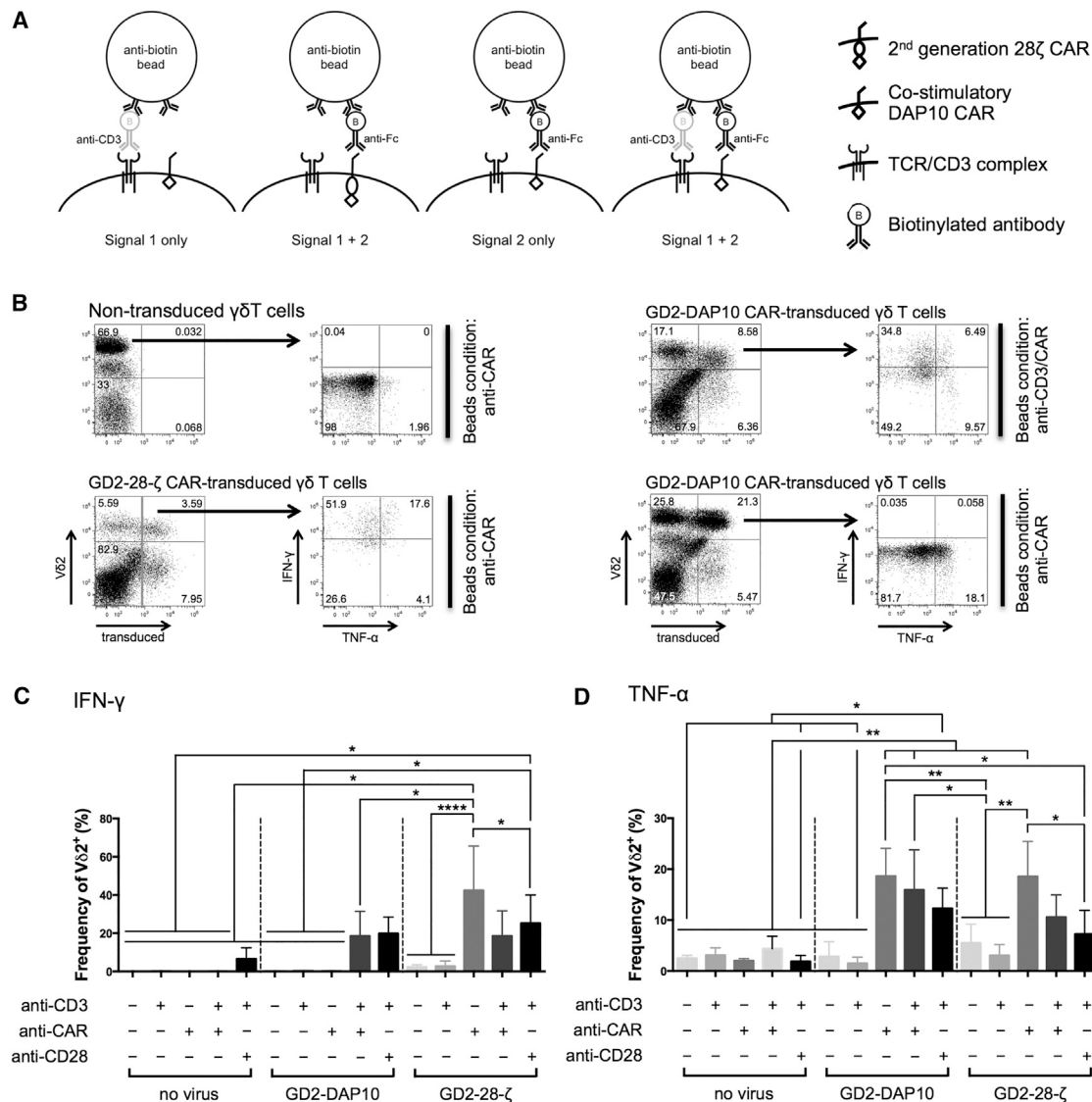


Figure 3. CAR-Mediated Co-stimulation Is Essential to Allow Functional Activation of $\gamma\delta$ T Cells upon TCR Engagement

(A) Schematic representation of the experimental design. Anti-CD3 antibodies coupled to the surface of beads were used to activate TCR (signal 1); anti-CAR (anti-Fc) antibodies activate the CAR (signal 2) through binding the CH2CH3 linker of the CAR. (B) Representative dot plots of $\gamma\delta$ T cell analysis by flow cytometry after incubation with beads labeled with anti-CD3 and/or anti-CAR antibodies. The binding and crosslinking of a second-generation CAR is sufficient to allow full T cell activation, since both signals 1 and 2 are provided by the CAR, i.e., by the CD3 ζ chain and the CD28 endodomain, respectively. According to the co-stimulation-only CAR concept, the engagement of a CAR providing only signal 2, e.g., by a DAP10 endodomain, does not lead to $\gamma\delta$ T cell activation. Only if signal 1, e.g., due to TCR activation, is provided in parallel $\gamma\delta$ T cells become activated and respond by functional changes, e.g., IFN- γ production. (C) The engagement of a second-generation CAR, i.e., GD2-28- ζ , is sufficient to induce a profound production of pro-inflammatory IFN- γ by $\gamma\delta$ T cells. In contrast, $\gamma\delta$ T cells expressing a co-stimulation-only CAR (GD2-DAP10) produce IFN- γ only upon both TCR and CAR engagement. (D) No differences in TNF- α production were found for respective anti-CD3 and/or anti-CAR antibody-stimulated $\gamma\delta$ T cells expressing the GD2-DAP10 or the GD2-28- ζ CAR. For (C) and (D), we performed Bonferroni post hoc analysis after one-way ANOVA. Graphs display merged results from two independent experiments ($n = 4$ –6 independent donors). Error bars represent mean values $\pm$ SD.

of either CAR promoted a more exhausted phenotype, which was associated with greater PD-1 expression in CAR⁺ T cells (Figures 7B and 7C; Figure S3). After 10 days of culture (5 days after transduction), no differences were seen in the percentages of exhausted cells between the two CAR⁺ groups (Figure 7C), but, by day 16 (11 days

after transduction), the GD2-DAP10 CAR⁺ population contained significantly fewer exhausted cells than the GD2-28- ζ CAR⁺ population and no difference in exhaustion markers compared with non-transduced cells (Figure 7D). This tendency was also seen at day 24 (Figure S4). Moreover, comparison of frequencies of activated events,

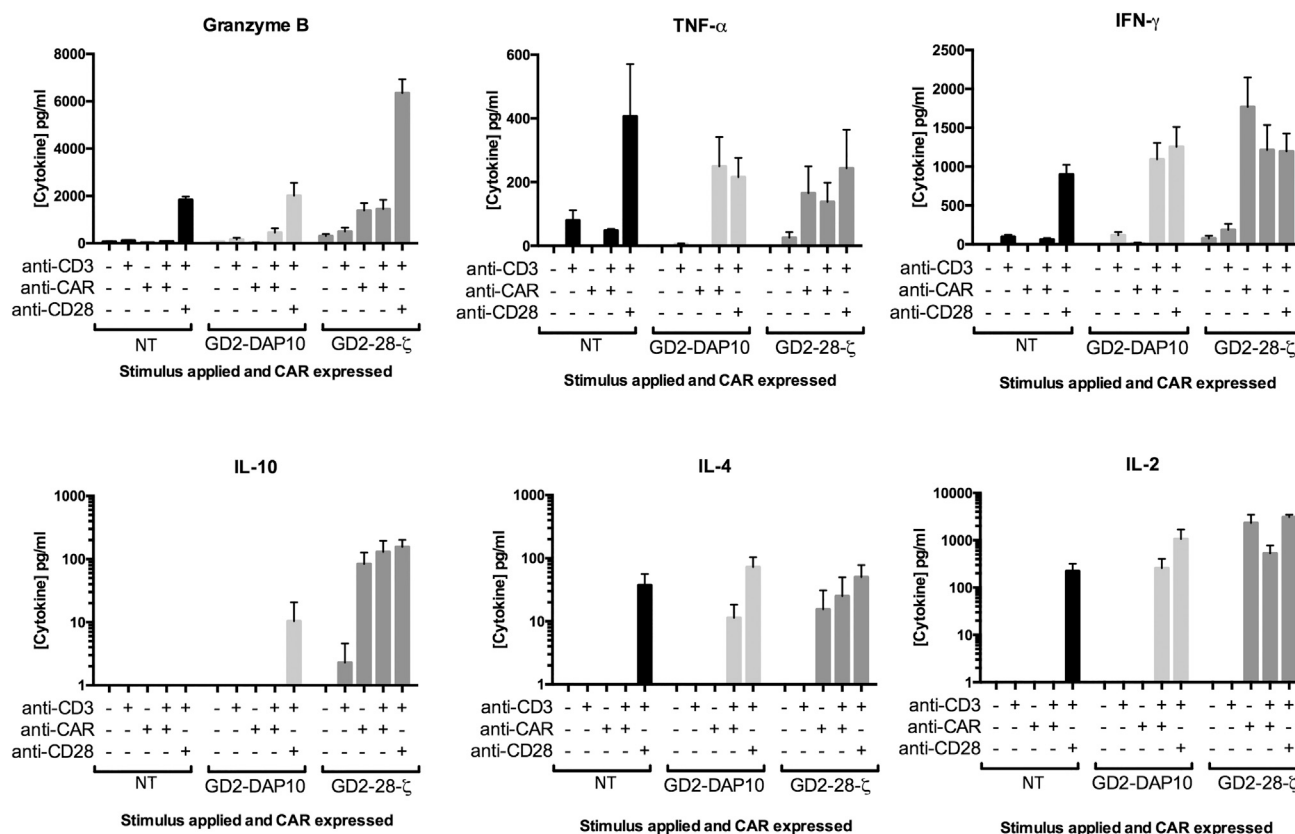


Figure 4. Cytokine Secretion by GD2-DAP10 Vδ2⁺ Cells Is Dependent on CD3 and CAR Engagement

Cytokine production by Vδ2⁺ γδT cells expressing GD2-DAP10 or GD2-28-ζ CARs following 23 hr stimulation with antibody-coated beads engaging CD3, the CAR, or both. Non-transduced γδT cells (NT) are included for comparison, and CD3/CD28 bead stimulation was included as a positive control. Both the CAR and CD3 must be stimulated in order for GD2-DAP10 Vδ2⁺ γδT cells to produce cytokine, whereas CAR stimulation alone is sufficient to generate cytokine production by GD2-28-ζ Vδ2⁺ cells. Error bars indicate SEM of three independent donors.

i.e., cells that express activation markers CD25 and/or CD69, indicated a tendency toward a more activated phenotype (CD69⁺/CD25⁺) in GD2-28-ζ CAR⁺ γδT cells compared to GD2-DAP10 CAR⁺ γδT cells, in the absence of further antigenic stimulus (Figure S5). Thus, while CAR expression appears to impart a degree of exhaustion, it appears that this effect is less pronounced when the GD2-DAP10 CAR is expressed compared to the GD2-28-ζ CAR, perhaps related to lower levels of background activation in GD2-DAP10 CAR⁺ γδT cells.

DISCUSSION

On-target off-tumor toxicity is one of the major impediments to successful cancer immunotherapy. It is typically approached in two ways: either by avoiding antigens, which may be expressed on healthy tissues, restricting antigen choice, or by accepting non-tumor antigen expression and the potentially severe toxicities that accompany it. Examples of this approach include anti-CD19 and anti-CD20 immunotherapies, which have been used against lymphoid malignancies.^{33,34} The elimination of healthy CD19⁺ B cells is considered to be an acceptable toxicity in this instance, as patients can be supported

long-term with intravenous immunoglobulin infusions as required. Neurotoxicity associated with the CD19 CAR T cells is an emerging concern.^{2,3,35} In other cases, such as neuroblastoma targeted with anti-GD2 monoclonal antibodies, the toxicity is more severe with pain being experienced presumably secondary to engagement of the GD2 expressed on peripheral nerve fibers. Fatal off-target toxicity has also been reported using CAR T cells targeting EGFR.⁷ The ability to differentiate between X-on tumor and X-on healthy would solve this problem, not only reducing toxicity but also making previously overlooked antigens available for targeting.

Here we have demonstrated that by providing supplemental DAP10 signals via a co-stimulation-only CAR, Vδ2⁺ γδT cell anti-neuroblastoma cytotoxicity can be induced. Neuroblastoma and DAP10 were chosen for this model because of low background killing of neuroblastoma by unmanipulated Vδ2⁺ γδT cells and because of previously documented immune evasion of NKG2D/DAP10 signaling in this disease.²² In tumor systems that do not evade killing through downregulation of membrane-bound or production of soluble NKG2D ligands, supplementing DAP10 signals to tumor-targeting T cells may not be of

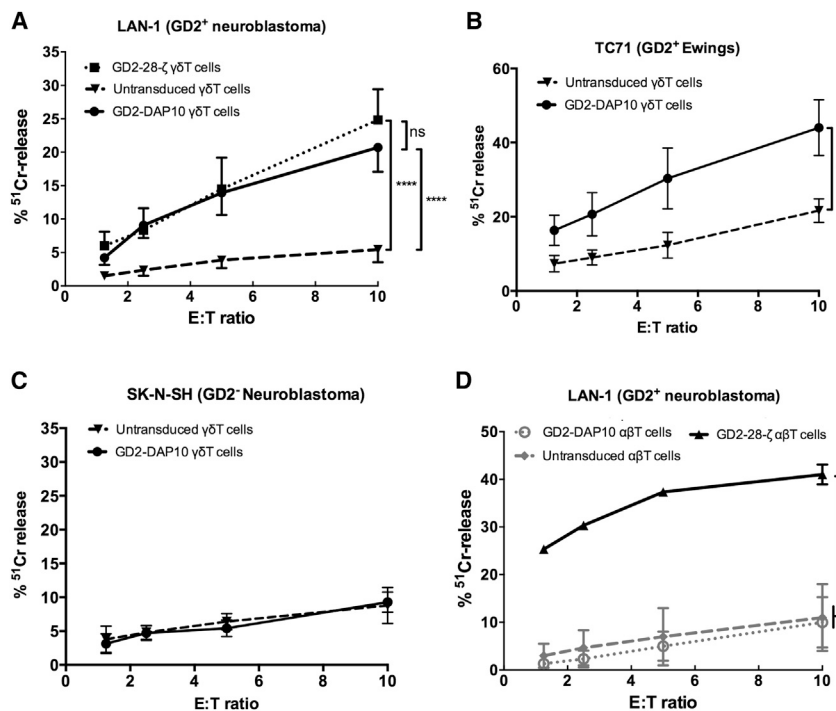


Figure 5. Vδ2⁺ γδT Cells Expressing GD2-DAP10 CARs Kill GD2⁺ Tumor Cells

(A and B) Specific killing of (A) neuroblastoma line LAN-1 ($n = 14$ independent donors for untransduced and (B) GD2⁺ Ewing sarcoma cell line TC-71 ($n = 3$ independent donors) by GD2-DAP10 CAR⁺. (C) This effect is not seen against the GD2⁻ neuroblastoma cell line SK-N-SH ($n = 3$ independent donors). (D) The GD2-DAP10 CAR does not function in αβT cells where signal 1 is absent, indicating that the signal is restricted to the co-stimulatory pathway ($n = 3$ independent donors for each subgroup). Frequencies are shown as mean $\pm$ SEM; significance was determined by Bonferroni post hoc correction after two-way ANOVA, with only E:T ratio 10:1 annotated for clarity. Cytotoxicity assays were carried out at day 6 after transduction of γδT cells.

benefit. Anti-tumor cytotoxicity is associated with production of IFN- γ , which is not produced by GD2-DAP10 CAR⁺ Vδ2 when the CAR or the γδTCR is stimulated in isolation (Figures 3 and 4). The lack of IFN- γ production in the absence of a second signal in the neuroblastoma model is consistent with our previously published findings, which demonstrated the need to coat neuroblastoma cells with antibodies that sensitize to ADCC to generate an optimal IFN- γ response in Vδ2 T cells.²⁰ Future characterization of other co-stimulatory endodomains within the co-stimulation-only CAR architecture will explain if intracellular detection of TNF- α , upon engagement of the GD2-DAP10 CAR without TCR engagement, is due to the endodomain chosen here, i.e., DAP10, which has previously been associated with TNF- α production after NKG2D activation.³⁶

We have demonstrated that the GD2-DAP10 CAR can induce Vδ2⁺ γδT cell cytotoxicity only against cell lines that engage both the γδTCR and the CAR, providing more precise control than conventional CD3ζ-containing CARs, which induce cytotoxicity in the presence of target antigen alone. This precision of control is robust in cell-based and bead-based assays, indicating mechanistic consistency. Unfortunately, due to the restriction of human GD2 expression to primary neurons, it is not feasible to perform cytotoxicity assays against healthy GD2⁺ cells. We have previously demonstrated that peripherally administered Vδ2⁺ γδT cells persist in an immunodeficient mouse model, retaining the ability to invade and kill subcutaneous neuroblastoma.¹⁹ However, it would not be possible to generate a murine model of healthy neuronal tissue that would have any possibility of engaging the human Vγ9Vδ2 TCR due to the species specificity of this interaction, precluding an in vivo demonstration

of the avoidance of on-target off-tumor toxicity. Furthermore, the majority of γδT cells that penetrate epithelial surfaces are of non-Vδ2⁺ subsets, which have greater antibody-independent anti-tumor ability²⁰ and, therefore, are less amenable to precise control. Vδ2⁺ γδT cells are generally restricted to the bloodstream and are much less abundant in the tissue space, providing exciting opportunities for targeting non-lymphoid hematological malignancies where immunotherapeutic approaches have been limited by toxicity.

Adoptively transferred cells used in cancer immunotherapy must be able to survive in the host and mount a secondary expansion in response to their target antigen. T cell exhaustion due to over-stimulation or tonic CAR signaling can render a cellular immunotherapy short lived or ineffective.^{32,37} The choice of scFv influences the tendency toward exhaustion; for example, the anti-GD2 scFv 14G2a generates more exhaustion than the huk666 scFv, most likely due to low levels of tonic signaling³² following scFv cross-linking. There is no sustained significant difference in the exhaustion phenotype of co-stimulation-only CAR Vδ2⁺ γδT cells and those transduced to express conventional second-generation CARs, though there is a tendency toward a less exhausted phenotype. While GD2-DAP10 Vδ2⁺ cells are able to produce IL-2 in response to simultaneous CD3 and CAR stimulation (Figure 4), the secondary expansion seen after re-challenge with target antigen-expressing tumor cells is modest. These experiments were performed 21 days after transduction in order to minimize background expansion kinetics, and they suggest that a further stimulus from an additional co-stimulatory endodomain may be of benefit. It is perhaps unfair to compare the secondary expansion potential of Vδ2 γδT cells with that of CAR-transduced αβT cells, however, as the baseline dynamics of these cell types are different. New light will be shed on the applicability of γδT cell subsets in future immunotherapy approaches, as novel insights underpin the important role of specific tissues in modeling their local γδT cell compartments.³⁸

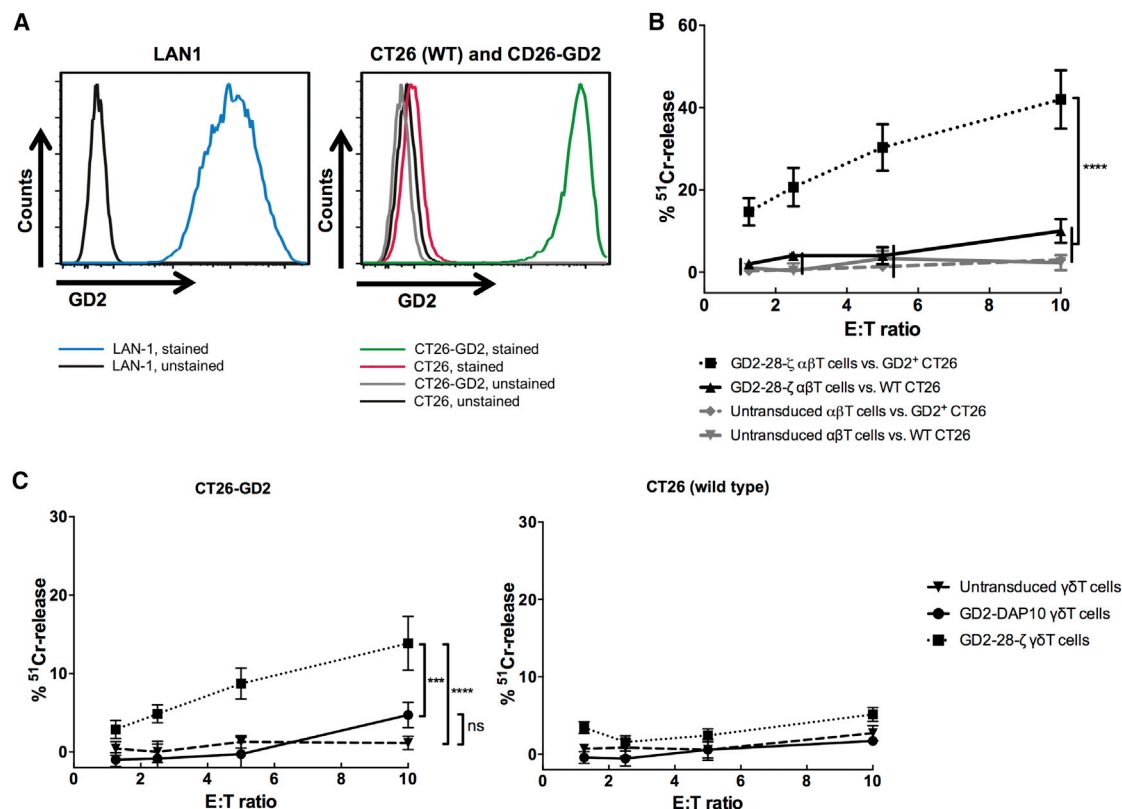


Figure 6. Target Cells Not Engaging the V γ 9V δ 2 Are Not Killed by GD2-DAP10 CAR V δ 2 $^{+}$ $\gamma\delta$ T Cells

(A) The murine colonic carcinoma cell line CT26 was transduced to express human GD2 and to derive the cell line CT26-GD2. Expression of GD2 was as homogeneous as that detected for LAN-1. (B) ^{51}Cr release assay demonstrating the ability of $\alpha\beta$ T cells expressing the GD2-28- ζ CAR to kill CT26-GD2 cells in an antigen-specific manner. Killing of wild-type CT26 and background killing by untransduced $\alpha\beta$ T cells are minimal ($n = 3$ independent donors for each subgroup). (C) ^{51}Cr release assay demonstrating that V δ 2 $^{+}$ $\gamma\delta$ T cells expressing the GD2-28- ζ CAR kill CT26-GD2 cells in an antigen-specific manner, even though CT26 does not engage the V δ 2 $^{+}$ $\gamma\delta$ TCR, whereas V δ 2 $^{+}$ $\gamma\delta$ T cells expressing the GD2-DAP10 CAR do not ($n = 7$ independent donors for each subgroup). A difference for killing of CT26 target cells was also determined at an E:T ratio of 10:1 using GD2-28- ζ CAR $^{+}$ cells compared to GD2-DAP10 CAR $^{+}$ cells ($n = 7$ independent donors for each subgroup). Frequencies are shown as mean $\pm$ SEM, with significance by Bonferroni post hoc correction after two-way ANOVA; only E:T ratio 10:1 is annotated for clarity.

While a direct route to translation is beyond the scope of this paper, and significant hurdles remain before co-stimulation-only CAR- $\gamma\delta$ T cell products become clinically available, the data suggest that they are worth investigating as an alternative to conventional CAR- $\alpha\beta$ T cell products. The addition of a co-stimulatory-only CAR may boost the efficacy without adding toxicity, something that is not possible using $\alpha\beta$ T CARs targeting single antigens due to the requirement for a CD3 ζ endodomain. Modest secondary expansion following antigen challenge may reduce the longevity of a product in a patient, but this need not limit efficacy if sufficient cells are administered at the outset. Autologous $\gamma\delta$ T cells can be expanded to sufficient numbers from the blood of cancer patients to have therapeutic efficacy against a range of tumor targets,^{39,40} and CAR-expressing $\gamma\delta$ T cells are readily expanded.⁴¹ We also demonstrate this (Figure 2D); extrapolation of these figures indicates that a cell dose of 10^7 – 10^8 cells would be achievable from a 100-mL blood draw.

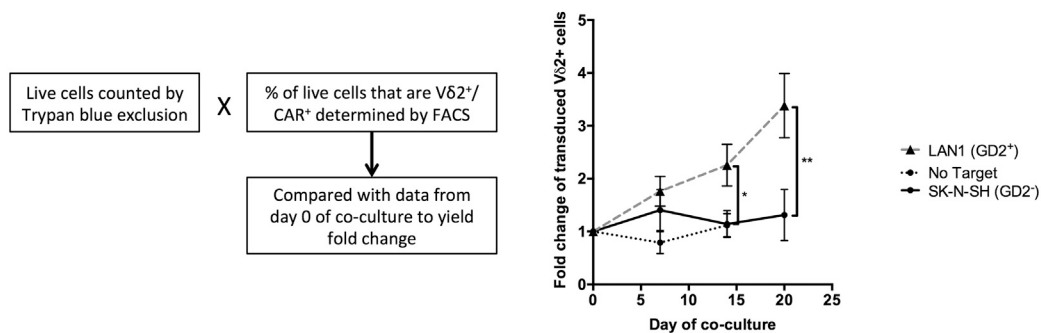
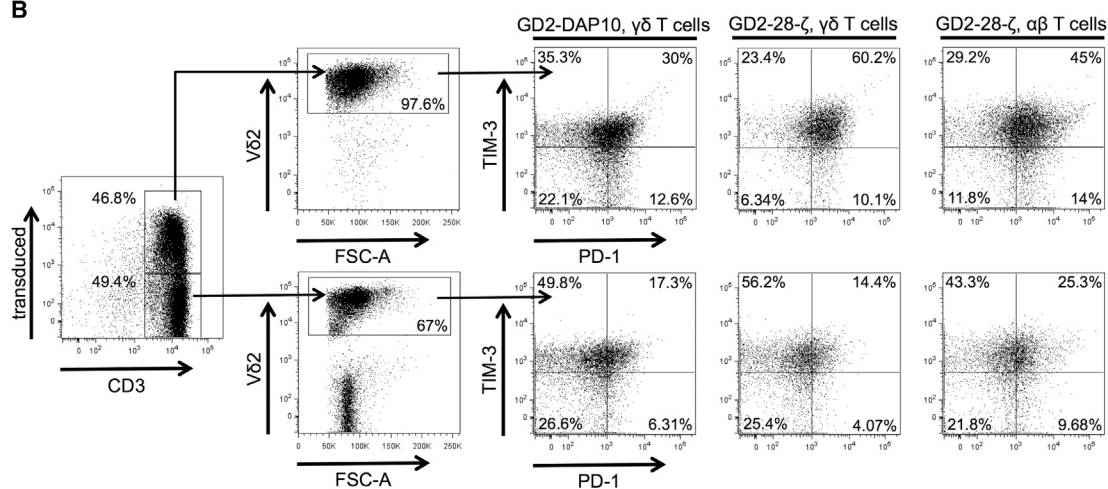
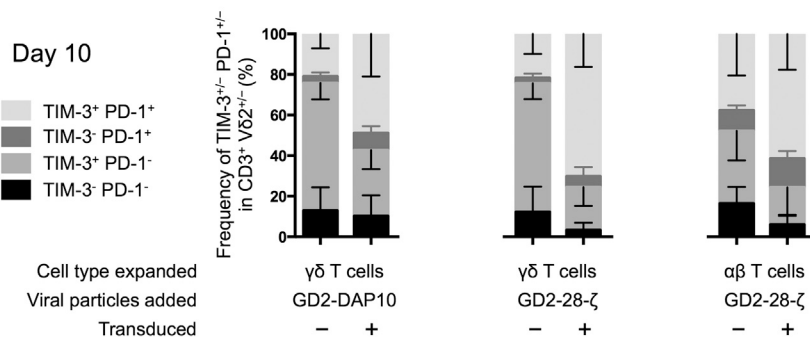
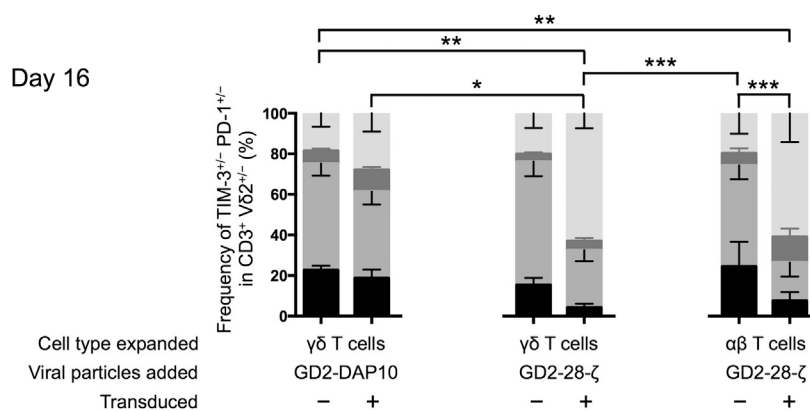
By introducing a novel, more precise approach to cellular cancer immunotherapy, co-stimulation-only CARs expressed in V δ 2 $^{+}$ $\gamma\delta$ T

cells present a choice. V δ 2 $^{+}$ $\gamma\delta$ T cells are less abundant, harder to expand, and less well understood than $\alpha\beta$ T cells, but they offer a means of specifically targeting tumors without harming healthy antigen-positive cells. The balance between force and precision of the anti-tumor response will depend on the context, but this approach at least offers an opportunity to target abundant and immunogenic molecules that were previously placed out of bounds by toxicity.

MATERIALS AND METHODS

Isolation of Peripheral Blood Mononuclear Cells and T Cell Expansion

Whole blood (20 mL) was diluted with 10 mL PBS + 500 μL 100 mM EDTA and layered on 20 mL Percoll. Interface peripheral blood mononuclear cells (PBMCs; 20 min, 300 $\times$ g, room temperature [RT]) were washed in PBS and re-suspended in 25 mL T cell medium (RPMI-1640 medium supplemented with L-glutamine [2 mM, Sigma-Aldrich], penicillin/streptomycin [100 IU/mL penicillin and 100 $\mu\text{g}/\text{mL}$ streptomycin, Sigma-Aldrich], and 10% fetal calf serum [FCS, v/v; Gibco]). In some experiments CD56 $^{+}$ cells were depleted

A**B****C****D**

(legend on next page)

from PBMC preparations (day 1) by positive selection using CD56 MicroBeads (Miltenyi Biotec). Depletions to less than 3% of CD56⁺ cells were achieved regularly. For $\gamma\delta$ T cell expansion, 5 μ M zoledronic acid (Actavis) and 100 IU/mL IL-2 (Aldesleukin, Novartis) were added to PBMC suspension after PBMC isolation (day 1). IL-2 was replenished every 2–3 days by removing half of the media from the well and replacing with fresh media containing 200 IU/mL IL-2. $\alpha\beta$ T cell expansion was induced by the addition of anti-CD3/CD28 Dynabeads (Gibco).

Construction of Retroviral Constructs

The retroviral vector used in all constructs was the oncoretroviral vector SFG,²⁶ pseudotyped with an RD114 envelope. DNA fragments were amplified using the Phusion HT II polymerase according to the manufacturer's instructions (Thermo Scientific). PCR was carried out in a PTC-200 DNA Engine (MJ Research). PCR products were extracted from 1% agarose gels using the Wizard SV Gel & PCR Clean-Up kit (Promega). Sample concentrations were determined using a NanoDrop ND-1000 spectrophotometer (Thermo Scientific). The CAR ectodomain comprised an scFv against human GD2 (clone huk666), and a spacer derived from the human IgG4 CH2-CH3 portion has been described before in third-generation format.²⁷ The endodomain was generated using oligo-assembly PCR based on a codon-optimized sequence of the human DAP10 endodomain (NCBI Gene ID 10870). In addition to the CAR construct, RQR8, which is a marker bearing a CD34 epitope,²⁸ was included, separated from the CAR by a cleavable 2A peptide. This allows CAR-expressing cells to be detected by flow cytometry by staining using the anti-CD34 antibody clone QBend10.

Production of Viral Particles by Transfection

The 293T cells (1.5×10^6 /100 mm² dish; Nunclon Delta Surface, Thermo Fisher Scientific) were plated at day 1 in 293T medium (DMEM, 10% FCS, v/v). The γ -retroviral particles were produced by co-transfection of 293T cells at day 2 using GeneJuice Transfection Reagent (Novagen/Millipore), in accordance with the manufacturer's directions. Viral particle-containing supernatants were harvested at day 4; medium was replenished and harvested at day 5. Supernatants were pooled, filtered (0.45- μ m filter, Millipore), and directly used for transductions or stored at -80°C .

Transduction of T Cells

Transduction of T cells was carried out in Retronectin (Takara Bio)-coated 24-well plates, which were pre-loaded with viral supernatant. The 0.5×10^6 T cells suspended in 0.5 mL T cell medium + 400 IU IL-2 were combined with 1.5 mL viral supernatant and centrifuged for 40 min ($1,000 \times g$ at RT). Typically, 12×10^6 T cells per donor were plated for transduction. Timing of transduction differed between $\gamma\delta$ T cells and $\alpha\beta$ T cells due to expansion dynamics. $\gamma\delta$ T cell expansion was stimulated with 5 μ M zoledronic acid (Actavis) and 100 IU/mL IL-2 (Aldesleukin) and transduction performed at day 5. At day 8 of culture (day 3 after transduction), cells were pooled, washed, and plated at 2×10^6 cells/mL in T cell medium + 100 IU IL-2/mL (24-well plates, Nunclon Delta Surface, Thermo Scientific). Transduction efficiency was determined by flow cytometry at day 10 (day 5 after transduction). In the case of $\alpha\beta$ T cells, transduction was performed 72 hr following stimulation with anti-CD3/CD28 Dynabeads (Gibco) in the presence of 100 IU/mL IL-2 (Aldesleukin), with re-plating 3 days after transduction. Cytotoxicity assays were performed 6 days post-transduction in both cases.

Flow Cytometry

For the analysis of intracellular cytokines, cells were incubated for 4 hr in T cell medium supplemented with $1 \times 5 \mu\text{g/mL}$ Brefeldin A (BioLegend). For exclusion of dead cells, the Zombie Yellow Fixable Viability Kit (BioLegend) was used. For Fc-receptor blocking, cells were incubated in PBS supplemented with FcR Blocking Reagent (Miltenyi Biotec). For intracellular staining of cytokines, the Inside Stain Kit (Miltenyi Biotec) was used. For absolute cell counting in secondary expansion experiments, CountBright beads (Molecular Probes/Life Technologies) were applied. The following antibodies were used to perform analysis by flow cytometry: anti-CD3-biotin (OKT3), anti-CD3-BV785 (OKT3), anti-CD3-APC (UCHT1), anti-CD56-FITC (HCD56), anti-CD69-APC-Cy (FN50), anti-CD279/PD-1-APC-Cy7 (EH12.2H7), anti-CD366/Tim-3-PE-Cy7 (F38-2E2), anti-TCR V δ 2-PE (B6), anti-TCR $\alpha\beta$ -APC (IP26), anti-IFN- γ -FITC (4S.B3), anti-TNF- α -BV711 (MAB11), anti-GD2-PE (14G2a) from BioLegend, anti-CD34-APC (QBend10) from R&D Systems, and anti-CD25-FITC (M-A251) from BD Pharmingen.

Figure 7. Expression of GD2-DAP10 CAR in V δ 2⁺ $\gamma\delta$ T Cells Leads to Less Exhaustion and Activation Marker Expression Than GD2-28- ζ CAR

(A) V δ 2⁺ T cells expressing the GD2-DAP10 CAR were transduced and then maintained in medium containing 100 IU/mL IL-2 for 3 weeks. They were then washed thoroughly and co-cultured with irradiated GD2⁺ LAN-1 or GD2⁺ SK-N-SH neuroblastoma cells in the absence of exogenous IL-2. A combination of live cell counting by trypan blue exclusion and fluorescence-activated cell sorting (FACS) analysis for CAR⁺/V δ 2⁺ cells was used to determine fold change over a 3-week co-culture. Only in the presence of GD2⁺ targets did the CAR⁺ cells mount a secondary expansion ($n = 3$ independent donors, fold change compared by paired t test at day 14 and day 20). (B) Gating to determine phenotype of CAR transduced $\gamma\delta$ T and $\alpha\beta$ T cells over prolonged culture in medium containing 100 IU/mL IL-2. Transduction efficiencies as well as the expression of TIM-3 and PD-1 were assessed by flow cytometry at each time point. Note that analysis of TIM-3 and PD-1 expression for conventional $\alpha\beta$ T cells was based on gating on V δ 2⁺ events. Cells expressing TIM-3 and/or PD-1 were considered to have an exhausted phenotype. (C) After 10 days of culture (5 days after transduction), no significant differences were found by comparing the frequencies of exhausted events in all subgroups. Exhausted cells were defined as those expressing at least one of the exhaustion markers TIM-3 or PD-1. (D) At day 16 of culture (10 days after transduction), $\alpha\beta$ T or $\gamma\delta$ T cells expressing GD2-28- ζ CARs were more exhausted than either untransduced $\alpha\beta$ T or $\gamma\delta$ T cells. Furthermore, $\gamma\delta$ T cells expressing the GD2-28- ζ CAR were significantly more exhausted than $\gamma\delta$ T cells expressing the GD2-DAP10 CAR. As in (C), for the purposes of this analysis, cells were considered to be exhausted if they expressed TIM-3, PD-1, or both. (C) and (D) display combined results from two independent experiments ($n = 5$ for $\gamma\delta$ T cells and $n = 6$ for $\alpha\beta$ T cells) subjected to Bonferroni post hoc analysis after one-way ANOVA. Columns represent mean values $\pm$ SD.

⁵¹Cr Killing Assay

To assess the killing capacity of T cells, 5,000 ⁵¹Cr-labeled target cells were co-cultured with 50,000, 25,000, 12,500, or 6,250 effector cells in 200 μ L T cell medium + 100 IU IL-2/mL. Cells were incubated in 96-well V-bottomed plates (Greiner Bio-one) for 4 hr. 50 μ L supernatant was transferred into Isoplate-96 HB plates (PerkinElmer). After the addition of 150 μ L scintillation cocktail (Optiphase Supermix, PerkinElmer), samples were incubated overnight at RT. Counts were acquired using a 1450 MicroBeta TriLux scintillation counter (PerkinElmer).

Activation of T Cells Using Antibody-Coated Beads

Anti-Biotin MACS iBeads (Miltenyi Biotec) were labeled with 10 μ g anti-CD3-biotin (OKT3, BioLegend), 10 μ g anti-IgG (Fc)-biotin (Novex/Life Technologies), or 10 μ g anti-CD3-biotin and 10 μ g anti-IgG (Fc)-biotin, respectively. After incubation (10 min, 4°C), beads were washed twice with PBS and re-suspended in T cell medium. To assess T cell functionality and act as a positive control, anti-CD3/CD28 Dynabeads were re-suspended in T cell medium as well. Bead suspensions (100 μ L) were plated in 96-well U-bottom plates (Thermo Scientific). Effector T cell preparations were added (0.2×10^6 cells/well in 100 μ L T cell medium) and incubated with beads for 24 hr. No exogenous IL-2 was added to the medium in either case. After 23 hr, 100 μ L supernatant was harvested for analysis of cytokine secretion by Cytometric Bead Array (CBA) and intracellular cytokine staining, i.e., data generated by CBA and intracellular cytokine staining are from the same experiments and are based on the same sample size.

Cytokine Secretion Assay

The BD CBA Flex Set Cytometric Bead Array (Becton Dickinson) was used in accordance with the manufacturer's instructions to measure the concentrations of cytokines, including IL-2, IL-4, IL-10, IL-17a, TNF- α , IFN- γ , and Granzyme B, in the supernatant of $\gamma\delta$ T cells stimulated for 23 hr with antibody-coated beads as above.

Secondary Expansion of Transduced Cells following Antigen Challenge

The $\gamma\delta$ T cells that had been transduced to express the GD2-DAP10 CAR were maintained for 3 weeks following transduction in T cell medium and 100 IU/mL IL-2 (Aldesleukin). Following the 3-week culture with no further TCR stimulus, the $\gamma\delta$ T cells were co-cultured with irradiated (60 Gy) neuroblastoma cell lines (LAN-1 or SK-N-SH) at a 1:1 effector:target (E:T) ratio. Live cell numbers were determined using Trypan blue exclusion, and the number of live TCRV δ 2⁺/CAR⁺ $\gamma\delta$ T cells was determined by flow cytometry using the Zombie Yellow fixable viability dye (BioLegend) in addition to staining for TCR V δ 2 and the CD34 epitope, which is co-expressed with the CAR. By multiplying the number of live cells by the percentage of live cells that were TCRV δ 2⁺/CAR⁺, the absolute cell number could be determined and used to calculate a fold change based on the starting cell number.

Cell Lines

Human neuroblastoma cell lines Kelly, SK-N-SH, and LAN-1 and the murine colonic carcinoma cell line CT26 were originally obtained

from the ATCC. The Ewing sarcoma cell line TC-71 was obtained from DSMZ. CT26-GD2 cells were generated by transducing wild-type CT26 with SFG gammaretrovirus encoding GD2 and GD3 synthase co-expressed via foot and mouth virus 2A self-cleaving peptide sequence, kindly provided by M. Pule (UCL). After transduction, single GD2-expressing clones were generated by limiting dilution and expanded.

Statistical Analysis

Analyses were performed using GraphPad Prism version 6.0. In instances where comparison was made between two groups with one variable only, a t test was used. Comparisons of more than two groups defined by one factor were done by one-way ANOVA followed by Bonferroni post hoc analysis. For comparisons of conditions where there was more than one variable (for example E:T ratio and transductions status), a two-way ANOVA with Bonferroni correction was used to determine statistical significance. These results were deemed to be statistically significant if $p < 0.05$. Where applicable, the following notation is used to denote significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$. Where numerical data are shown in the text, they are displayed as mean $\pm$ SEM unless otherwise stated.

SUPPLEMENTAL INFORMATION

Supplemental Information includes five figures and can be found with this article online at <http://dx.doi.org/10.1016/j.ymthe.2017.03.002>.

AUTHOR CONTRIBUTIONS

J.F. devised the initial concept and designed, performed, and analyzed experiments. P.A. conceived, designed, performed, and analyzed experiments. N.D.W.D. designed, performed, and analyzed experiments. B.F. optimized and performed Cytometric Bead Array experiments. A.C. optimized the transduction methods. G.W.-K.C. performed experiments. K.G. provided immunological support. J.A. designed experiments, supervised the work, and contributed to manuscript preparation in conjunction with J.F. and P.A.

CONFLICTS OF INTEREST

The authors have no conflicts of interest to declare.

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REFERENCES

- Chmielewski, M., Hombach, A.A., and Abken, H. (2013). Antigen-Specific T-Cell Activation Independently of the MHC: Chimeric Antigen Receptor-Redirected T Cells. *Front. Immunol.* 4, 371.
- Maude, S.L., Frey, N., Shaw, P.A., Aplenc, R., Barrett, D.M., Bunin, N.J., Chew, A., Gonzalez, V.E., Zheng, Z., Lacey, S.F., et al. (2014). Chimeric antigen receptor T cells for sustained remissions in leukemia. *N. Engl. J. Med.* 371, 1507–1517.
- Lee, D.W., Kochenderfer, J.N., Stetler-Stevenson, M., Cui, Y.K., Delbrook, C., Feldman, S.A., Fry, T.J., Orentas, R., Sabatino, M., Shah, N.N., et al. (2015). T cells expressing CD19 chimeric antigen receptors for acute lymphoblastic leukaemia in children and young adults: a phase 1 dose-escalation trial. *Lancet* 385, 517–528.
- Pule, M.A., Savoldo, B., Myers, G.D., Rossig, C., Russell, H.V., Dotti, G., Huls, M.H., Liu, E., Gee, A.P., Mei, Z., et al. (2008). Virus-specific T cells engineered to coexpress tumor-specific receptors: persistence and antitumor activity in individuals with neuroblastoma. *Nat. Med.* 14, 1264–1270.
- Louis, C.U., Savoldo, B., Dotti, G., Pule, M., Yvon, E., Myers, G.D., Rossig, C., Russell, H.V., Diouf, O., Liu, E., et al. (2011). Antitumor activity and long-term fate of chimeric antigen receptor-positive T cells in patients with neuroblastoma. *Blood* 118, 6050–6056.
- Lamers, C.H.J., Sleijfer, S., Vulto, A.G., Kruit, W.H.J., Kliffen, M., Debets, R., Gratama, J.W., Stoter, G., and Oosterwijk, E. (2006). Treatment of metastatic renal cell carcinoma with autologous T-lymphocytes genetically retargeted against carbonic anhydrase IX: first clinical experience. *J. Clin. Oncol.* 24, e20–e22.
- Morgan, R.A., Yang, J.C., Kitano, M., Dudley, M.E., Laurencot, C.M., and Rosenberg, S.A. (2010). Case report of a serious adverse event following the administration of T cells transduced with a chimeric antigen receptor recognizing ERBB2. *Mol. Ther.* 18, 843–851.
- Cheever, M.A., Allison, J.P., Ferris, A.S., Finn, O.J., Hastings, B.M., Hecht, T.T., Mellman, I., Prindiville, S.A., Viner, J.L., Weiner, L.M., and Matrisian, L.M. (2009). The prioritization of cancer antigens: a national cancer institute pilot project for the acceleration of translational research. *Clin. Cancer Res.* 15, 5323–5337.
- Yu, A.L., Gilman, A.L., Ozkaynak, M.F., London, W.B., Kreissman, S.G., Chen, H.X., Smith, M., Anderson, B., Villablanca, J.G., Matthay, K.K., et al.; Children's Oncology Group (2010). Anti-GD2 antibody with GM-CSF, interleukin-2, and isotretinoin for neuroblastoma. *N. Engl. J. Med.* 363, 1324–1334.
- Simon, T., Hero, B., Faldum, A., Handgretinger, R., Schrappe, M., Niethammer, D., and Berthold, F. (2004). Consolidation treatment with chimeric anti-GD2-antibody ch14.18 in children older than 1 year with metastatic neuroblastoma. *J. Clin. Oncol.* 22, 3549–3557.
- Hayday, A.C. (2009). Gammadelta T cells and the lymphoid stress-surveillance response. *Immunity* 31, 184–196.
- Vantourout, P., and Hayday, A. (2013). Six-of-the-best: unique contributions of $\gamma\delta$ T cells to immunology. *Nat. Rev. Immunol.* 13, 88–100.
- Poupot, M., and Fournié, J.-J. (2004). Non-peptide antigens activating human Vgamma9/Vdelta2 T lymphocytes. *Immunol. Lett.* 95, 129–138.
- Asslan, R., Pradines, A., Pratz, C., Allal, C., Favre, G., and Le Gaillard, F. (1999). Epidermal growth factor stimulates 3-hydroxy-3-methylglutaryl-coenzyme A reductase expression via the ErbB-2 pathway in human breast adenocarcinoma cells. *Biochem. Biophys. Res. Commun.* 260, 699–706.
- Vavassori, S., Kumar, A., Wan, G.S., Ramanjaneyulu, G.S., Cavallari, M., El Daker, S., Beddoe, T., Theodossis, A., Williams, N.K., Gostick, E., et al. (2013). Butyrophilin 3A1 binds phosphorylated antigens and stimulates human $\gamma\delta$ T cells. *Nat. Immunol.* 14, 908–916.
- Harly, C., Guillaume, Y., Nedellec, S., Peigné, C.-M., Mönkkönen, H., Mönkkönen, J., Li, J., Kuball, J., Adams, E.J., Netzer, S., et al. (2012). Key implication of CD277/butyrophilin-3 (BTN3A) in cellular stress sensing by a major human $\gamma\delta$ T-cell subset. *Blood* 120, 2269–2279.
- Sandstrom, A., Peigné, C.-M., Léger, A., Crooks, J.E., Konczak, F., Gesnel, M.-C., Breathnach, R., Bonneville, M., Scotet, E., and Adams, E.J. (2014). The intracellular B30.2 domain of butyrophilin 3A1 binds phosphoantigens to mediate activation of human V γ 9V δ 2 T cells. *Immunity* 40, 490–500.
- Silva-Santos, B., Serre, K., and Norell, H. (2015). $\gamma\delta$ T cells in cancer. *Nat. Rev. Immunol.* 15, 683–691.
- Fisher, J.P., Flutter, B., Wesemann, F., Frosch, J., Rossig, C., Gustafsson, K., and Anderson, J. (2015). Effective combination treatment of GD2-expressing neuroblastoma and Ewing's sarcoma using anti-GD2 ch14.18/CHO antibody with V γ 9V δ 2+ $\gamma\delta$ T cells. *Oncoimmunology* 5, e1025194.
- Fisher, J.P., Yan, M., Heuvelink, J., Carter, L., Abolhassani, A., Frosch, J., Wallace, R., Flutter, B., Capsomidis, A., Hubank, M., et al. (2014). Neuroblastoma killing properties of V δ 2 and V δ 2-negative $\gamma\delta$ T cells following expansion by artificial antigen-presenting cells. *Clin. Cancer Res.* 20, 5720–5732.
- Nedellec, S., Sabourin, C., Bonneville, M., and Scotet, E. (2010). NKG2D costimulates human V gamma 9V delta 2 T cell antitumor cytotoxicity through protein kinase C theta-dependent modulation of early TCR-induced calcium and transduction signals. *J. Immunol.* 185, 55–63.
- Raffaghello, L., Prigione, I., Airolidi, I., Camoriano, M., Levreri, I., Gambini, C., Pende, D., Steinle, A., Ferrone, S., and Pistoia, V. (2004). Downregulation and/or release of NKG2D ligands as immune evasion strategy of human neuroblastoma. *Neoplasia* 6, 558–568.
- Raffaghello, L., Prigione, I., Airolidi, I., Camoriano, M., Morandi, F., Bocca, P., Gambini, C., Ferrone, S., and Pistoia, V. (2005). Mechanisms of immune evasion of human neuroblastoma. *Cancer Lett.* 228, 155–161.
- Lanitis, E., Poussin, M., Klattenhoff, A.W., Song, D., Sandaltzopoulos, R., June, C.H., and Powell, D.J., Jr. (2013). Chimeric antigen receptor T Cells with dissociated signaling domains exhibit focused antitumor activity with reduced potential for toxicity in vivo. *Cancer Immunol. Res.* 1, 43–53.
- Kloss, C.C., Condomines, M., Cartellieri, M., Bachmann, M., and Sadelain, M. (2013). Combinatorial antigen recognition with balanced signaling promotes selective tumor eradication by engineered T cells. *Nat. Biotechnol.* 31, 71–75.
- Rivière, I., Brose, K., and Mulligan, R.C. (1995). Effects of retroviral vector design on expression of human adenosine deaminase in murine bone marrow transplant recipients engrafted with genetically modified cells. *Proc. Natl. Acad. Sci. USA* 92, 6733–6737.
- Thomas, S., Straathof, K., Himoudi, N., Anderson, J., and Pule, M. (2016). An Optimized GD2-Targeting Retroviral Cassette for More Potent and Safer Cellular Therapy of Neuroblastoma and Other Cancers. *PLoS One* 11, e0152196.
- Philip, B., Kokalaki, E., Mekkaoui, L., Thomas, S., Straathof, K., Flutter, B., Marin, V., Marafioti, T., Chakraverty, R., Linch, D., et al. (2014). A highly compact epitope-based marker/suicide gene for easier and safer T-cell therapy. *Blood* 124, 1277–1287.
- Kailayangiri, S., Altvater, B., Meltzer, J., Pscherer, S., Luecke, A., Dierkes, C., Titz, U., Leuchte, K., Landmeier, S., Hotfilder, M., et al. (2012). The ganglioside antigen G(D2) is surface-expressed in Ewing sarcoma and allows for MHC-independent immune targeting. *Br. J. Cancer* 106, 1123–1133.
- Kato, Y., Tanaka, Y., Tanaka, H., Yamashita, S., and Minato, N. (2003). Requirement of species-specific interactions for the activation of human gamma delta T cells by pamidronate. *J. Immunol.* 170, 3608–3613.
- Green, A.E., Lissina, A., Hutchinson, S.L., Hewitt, R.E., Temple, B., James, D., Boulter, J.M., Price, D.A., and Sewell, A.K. (2004). Recognition of nonpeptide antigens by human V γ 9V δ 2 T cells requires contact with cells of human origin. *Clin. Exp. Immunol.* 136, 472–482.
- Long, A.H., Haso, W.M., Shern, J.F., Wanhainen, K.M., Murgai, M., Ingaramo, M., Smith, J.P., Walker, A.J., Kohler, M.E., Venkateshwara, V.R., et al. (2015). 4-1BB costimulation ameliorates T cell exhaustion induced by tonic signaling of chimeric antigen receptors. *Nat. Med.* 21, 581–590.
- Jackson, H.J., Rafiq, S., and Brentjens, R.J. (2016). Driving CAR T-cells forward. *Nat. Rev. Clin. Oncol.* 13, 370–383.
- Batlevi, C.L., Matsuki, E., Brentjens, R.J., and Younes, A. (2016). Novel immunotherapies in lymphoid malignancies. *Nat. Rev. Clin. Oncol.* 13, 25–40.
- Davila, M.L., Riviere, I., Wang, X., Bartido, S., Park, J., Curran, K., Chung, S.S., Stefanski, J., Borquez-Ojeda, O., Olszewska, M., et al. (2014). Efficacy and toxicity management of 19-28z CAR T cell therapy in B cell acute lymphoblastic leukemia. *Sci. Transl. Med.* 6, 224ra25.

36. Rincon-Orozco, B., Kunzmann, V., Wrobel, P., Kabelitz, D., Steinle, A., and Herrmann, T. (2005). Activation of V gamma 9V delta 2 T cells by NKG2D. *J. Immunol.* *175*, 2144–2151.
37. Wherry, E.J., and Kurachi, M. (2015). Molecular and cellular insights into T cell exhaustion. *Nat. Rev. Immunol.* *15*, 486–499.
38. Di Marco Barros, R., Roberts, N.A., Dart, R.J., Vantourout, P., Jandke, A., Nussbaumer, O., Deban, L., Cipolat, S., Hart, R., Iannitto, M.L., et al. (2016). Epithelia Use Butyrophilin-like Molecules to Shape Organ-Specific $\gamma\delta$ T Cell Compartments. *Cell* *167*, 203–218.e17.
39. Fournié, J.J., Sicard, H., Poupot, M., Bezombes, C., Blanc, A., Romagné, F., Ysebaert, L., and Laurent, G. (2013). What lessons can be learned from $\gamma\delta$ T cell-based cancer immunotherapy trials? *Cell. Mol. Immunol.* *10*, 35–41.
40. Fisher, J.P.H., Heuijers, J., Yan, M., Gustafsson, K., and Anderson, J. (2014). $\gamma\delta$ T cells for cancer immunotherapy: A systematic review of clinical trials. *Oncoimmunology* *3*, e27572.
41. Deniger, D.C., Switzer, K., Mi, T., Maiti, S., Hurton, L., Singh, H., Huls, H., Olivares, S., Lee, D.A., Champlin, R.E., and Cooper, L.J. (2013). Bispecific T-cells expressing polyclonal repertoire of endogenous $\gamma\delta$ T-cell receptors and introduced CD19-specific chimeric antigen receptor. *Mol. Ther.* *21*, 638–647.